THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · HARD LEVEL CHAPTER TEST SET 3

Chapter 14: **Area** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A triangle has sides 48 cm, 55 cm and 73 cm. Its area is: 1

(a) 176 cm² **(b)** 1320 cm² **(c)** 1752 cm² **(d)** 2640 cm²

2. The perimeter of a rectangle is 140 cm and the length is 4 times the breadth. Its area is: 1

(a) 784 cm² **(b)** 1225 cm² **(c)** 756 cm² **(d)** 798 cm²

3. An isosceles trapezium has parallel sides 8 cm and 32 cm and non-parallel sides 20 cm. Its area is: 1

(a) 400 cm² **(b)** 330 cm² **(c)** 640 cm² **(d)** 320 cm²

4. The length of a rectangle is increased by 50% and its breadth decreased by 25%. If the original area is 800 cm², the new area (in cm²) is: 1

(a) 800 **(b)** 900 **(c)** 1200 **(d)** 1000

5. The area of a triangle is 150 cm². If the altitude on the side of 25 cm is 12 cm, the altitude on the side of 20 cm (in cm) is: 1

(a) 12 **(b)** 15 **(c)** 17 **(d)** 7

6. A rhombus has perimeter 328 cm and one diagonal 36 cm. Its area is: 1

(a) 2880 cm² **(b)** 1440 cm² **(c)** 2952 cm² **(d)** 5760 cm²

7. A 80 m by 44 m rectangular lawn has a path 3 m wide inside along its boundary. The area of the lawn left is: 1

(a) 3157 m² **(b)** 3484 m² **(c)** 2776 m² **(d)** 2812 m²

8. A triangle has sides 36 cm, 77 cm and 85 cm. Its area is: 1

(a) 2772 cm² **(b)** 1530 cm² **(c)** 1386 cm² **(d)** 198 cm²

9. **Assertion (A):** A rhombus with diagonals 10 cm and 24 cm has area 120 cm^2. 1

Reason (R): Area of a rhombus $= \frac{1}{2} \times d_1 \times d_2$.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. **Assertion (A):** A square of side 9 cm has area 81 cm^2. 1

Reason (R): The area of a triangle is half the product of its base and height.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. The area of a triangle is 176 cm² and one side is 22 cm. Find the altitude on that side and the area of a triangle with the same base and double the altitude. 2

12. The perimeter of a rectangle is 200 cm and its length exceeds its breadth by 30 cm. Find its area. 2

13. A rectangle's length is increased by 50% and breadth decreased by 40%. By what percent does the area change? (original area 100 units) 2

14. The area of a triangle is 255 cm² and one side is 34 cm. Find the altitude on that side and the area of a triangle with the same base and double the altitude. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. The perimeter of a rectangular plot is 144 m and its length is 3 times its breadth. Find its area and the cost of fencing at ₹ 25 per metre. 3

16. An isosceles trapezium has parallel sides 22 cm and 42 cm and each slant side 26 cm. Find its height and area. 3

17. A garden 45 m by 32 m has a rectangular pond 10 m by 10 m inside it and a path 2 m wide around the garden (outside). Find the area of the garden excluding the pond, and the area of the path. 3

OR

The perimeter of a rectangular plot is 216 m and its length is 5 times its breadth. Find its area and the cost of fencing at ₹ 8 per metre. 3

18. An isosceles trapezium has parallel sides 30 cm and 60 cm and each slant side 25 cm. Find its height and area. 3

OR

A garden 70 m by 50 m has a rectangular pond 20 m by 12 m inside it and a path 3 m wide around the garden (outside). Find the area of the garden excluding the pond, and the area of the path. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** An L-shaped hall is formed from a rectangular hall 12 m long and 8 m wide by cutting off a rectangular corner 5 m long and 4 m wide. The floor is to be tiled. 5

(i) Find the area of the full rectangle.
(ii) Find the area of the corner removed.
(iii) Find the area of the L-shaped hall.
(iv) Find the cost of flooring at ₹250 per m^2.

20. The area of a rhombus is 864 cm² and one diagonal is 36 cm. Find the other diagonal, the side, and the altitude. 5

OR

A trapezium ABCD has parallel sides AB = 22 cm and DC = 40 cm and height 14 cm. A parallelogram of base 22 cm and the same height is cut off from it, leaving a triangle. Find the area of the trapezium, the parallelogram and the remaining triangle. 5

END OF QUESTION PAPER — Total: 40 Marks